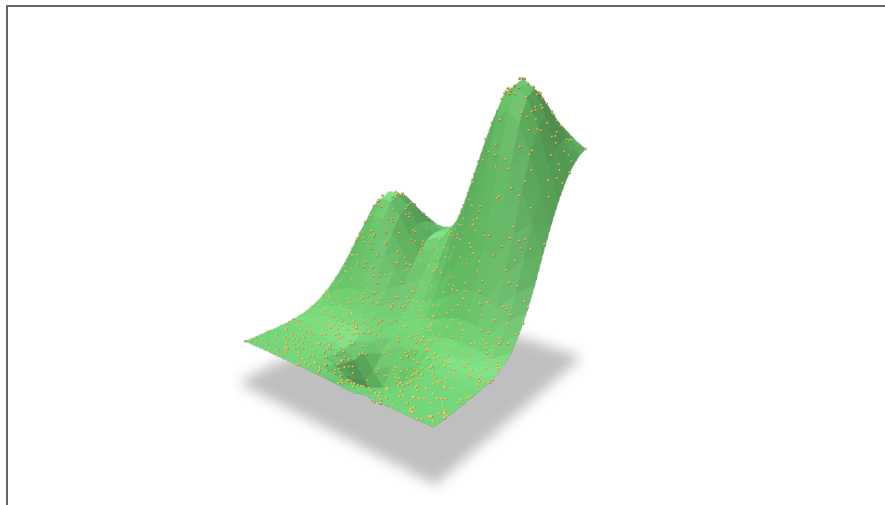
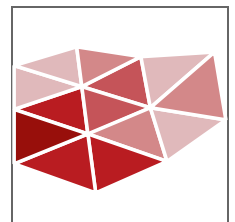


Parametric Surfaces

Exercise 2



Marc Alexa & Gabriel Dogadov





Exercise 2

General Information

Parametric surfaces from points

Moving Least Squares

Bézier Tensor Product Surfaces

You don't need to implement methods from linear algebra

Use a library: Eigen (C++), NumPy (Python), ...



Exercise 2

Schedule

Code presentation on 3rd of June

choose slot on ISIS (first come, first served)

available a few days in advance

Theory deadline also 3rd of June at 23:59



Exercise 2

Parametric Surfaces from points

Parametric surfaces:

$$\mathbf{q} : [a, b] \times [c, d] \rightarrow \mathbb{R}^3$$

In general:

$$\mathbf{q}(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \\ z(u, v) \end{pmatrix}$$

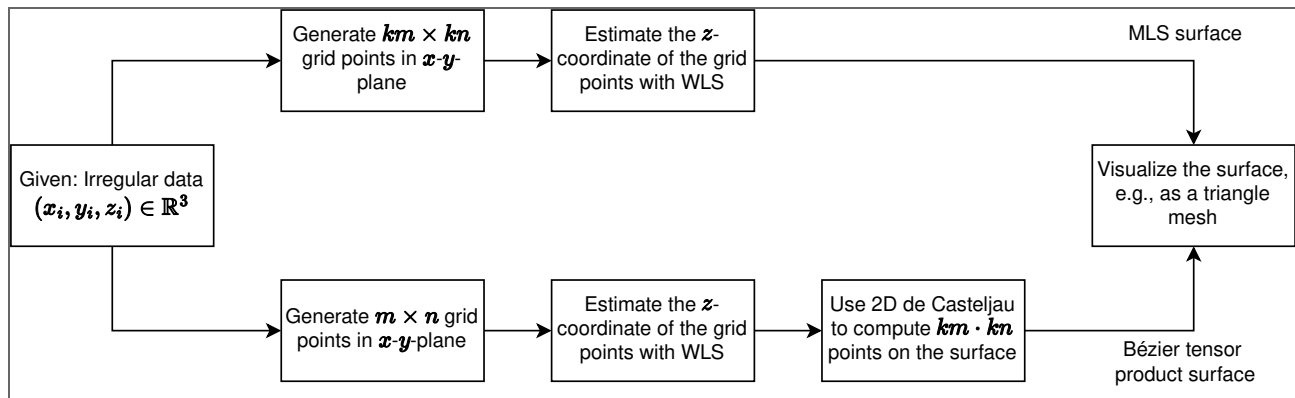
In this exercise as height field:

$$\mathbf{q}(u, v) = \begin{pmatrix} u \\ v \\ z(u, v) \end{pmatrix}$$



Exercise 2 Practice Part

Admap





Exercise 2 Practice Part

x 1

Load 3D points from an OFF file (already done, hopefully)

Compute a minimum bounding box (already done, hopefully)

Generate an $m \times n$ grid in the x - y -plane

The parameters m , n should be adjustable in the GUI

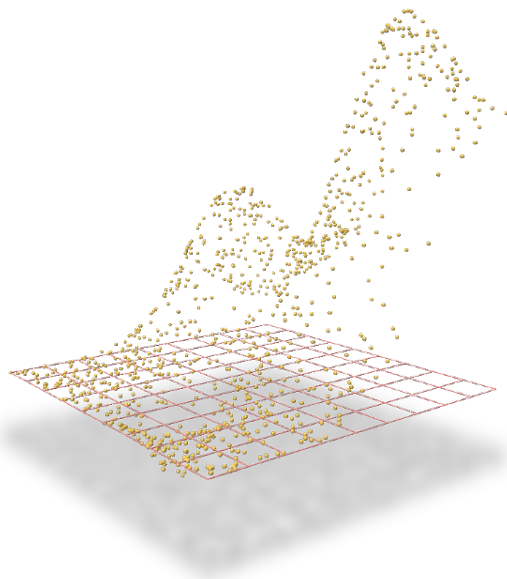
Estimate the z -values of the grid points using WLS

Visualize the grid points

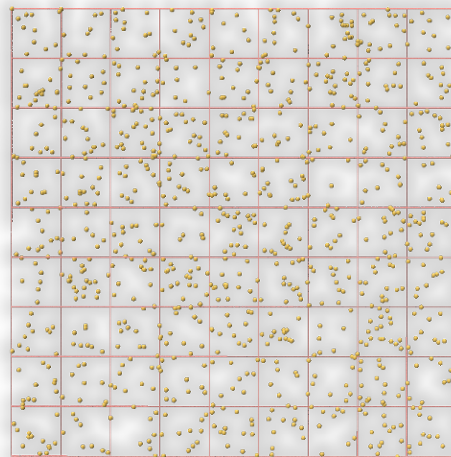
This task is a prerequisite for the other tasks!



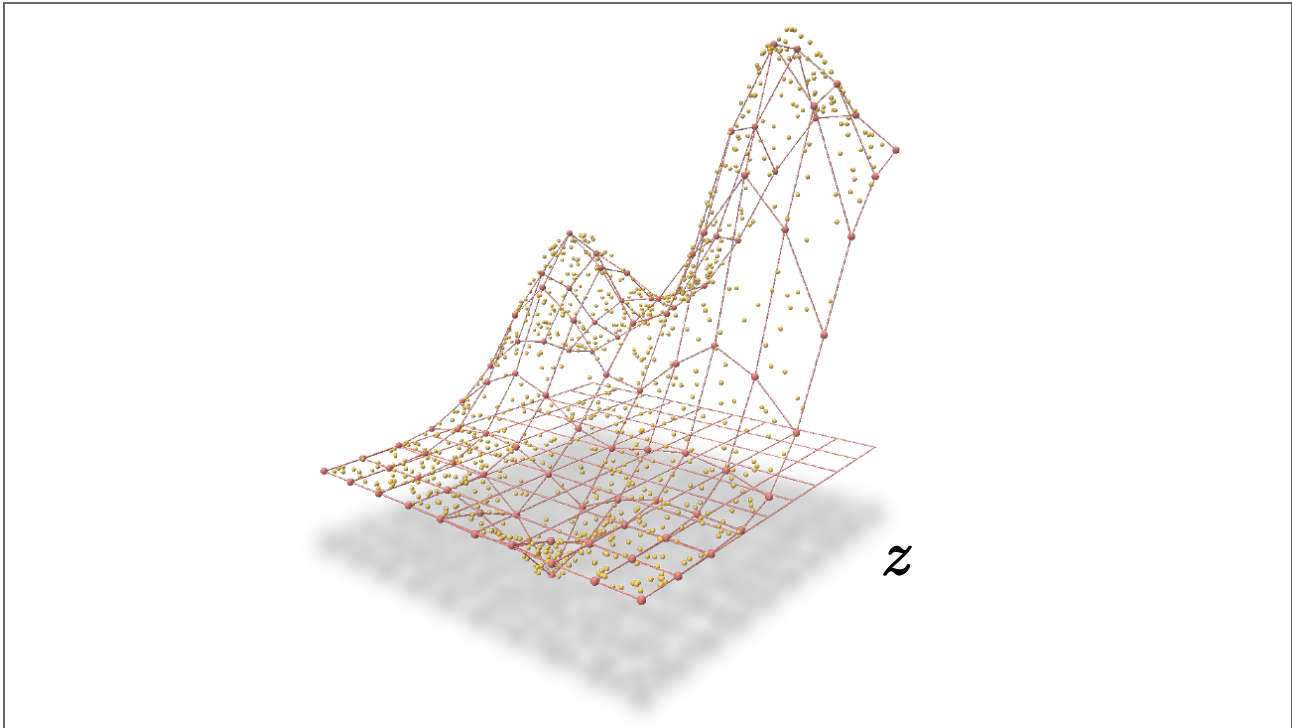
Exercise 2 Practice Part x 1 visualized



$m \times n$ grid (front view)



grid (top view)



Grid with WLS estimation of the z -coordinate



Exercise 2 Practice Part

Ex 1 - (Ordinary) Least Squares

Given: x and y coordinates (x_j, y_j) of new points
(the grid points)

Task: Estimate corresponding z coordinate z_j

Approximate $z_j = f(x_j, y_j)$ with $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ being
a bivariate quadratic polynomial

$$f(x, y) = s + r_x x + r_y y + q_{xx} x^2 + q_{xy} xy + q_{yy} y^2$$

$$= \begin{pmatrix} s & r_x & r_y & q_{xx} & q_{xy} & q_{yy} \end{pmatrix} \begin{pmatrix} 1 \\ x \\ y \\ x^2 \\ xy \\ y^2 \end{pmatrix}$$

The **parameters** are estimated from the data



Exercise 2 Practice Part x 1 - Linear System

System of linear equations:

$$z_i = f(x_i, y_i) = s + r_x x_i + r_y y_i + q_{xx} x_i^2 + q_{xy} x_i y_i + q_{yy} y_i^2$$

Initial data points (x_i, y_i, z_i) , $i \in \{0, \dots, n\}$

$$\begin{pmatrix} 1 & x_0 & y_0 & x_0^2 & x_0 y_0 & y_0^2 \\ 1 & x_1 & y_1 & x_1^2 & x_1 y_1 & y_1^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & y_n & x_n^2 & x_n y_n & y_n^2 \end{pmatrix} \begin{pmatrix} s \\ r_x \\ r_y \\ q_{xx} \\ q_{xy} \\ q_{yy} \end{pmatrix} = \begin{pmatrix} z_0 \\ z_1 \\ \vdots \\ z_n \end{pmatrix}$$



Exercise 2 Practice Part x 1 - Linear System II

The linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is overdetermined and typically not solvable $\Rightarrow$ solve it in the least-squares sense

$$\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 \quad \Leftrightarrow \quad \mathbf{A}^T \mathbf{A}\mathbf{x} = \mathbf{A}^T \mathbf{b}$$

Solve normal equation instead

Alternative: Suitable matrix decomposition



Exercise 2 Practice Part x 1 - Linear System III

Linear system of equations after applying the normal equation

$$\sum_{i=0}^n \begin{pmatrix} 1 & x_i & y_i & x_i^2 & x_i y_i & y_i^2 \\ x_i & x_i^2 & x_i y_i & x_i^3 & x_i^2 y_i & x_i y_i^2 \\ y_i & x_i y_i & y_i^2 & x_i^2 y_i & x_i y_i^2 & y_i^3 \\ x_i^2 & x_i^3 & x_i^2 y_i & x_i^4 & x_i^3 y_i & x_i^2 y_i^2 \\ x_i y_i & x_i^2 y_i & x_i y_i^2 & x_i^3 y_i & x_i^2 y_i^2 & x_i y_i^3 \\ y_i^2 & x_i y_i^2 & y_i^3 & x_i^2 y_i^2 & x_i y_i^3 & y_i^4 \end{pmatrix} \begin{pmatrix} s \\ r_x \\ r_y \\ q_{xx} \\ q_{xy} \\ q_{yy} \end{pmatrix} = \sum_{i=0}^n \begin{pmatrix} 1 \\ x_i \\ y_i \\ x_i^2 \\ x_i y_i \\ y_i^2 \end{pmatrix} z_i$$

Just a 6×6 matrix that is symmetric and positive semi-definite

Think about a suitable method for solving these kinds of systems



Exercise 2 Practice Part

Ex 1 - Weighted Least Squares

Data is often complex and heterogeneous

One "global" function for the entire data set often does not suffice

Weight the "influence" of the data points by their distance

$$\sum_{i=0}^n \theta(d_i) \begin{pmatrix} 1 & x_i & y_i & x_i^2 & x_i y_i & y_i^2 \\ x_i & x_i^2 & x_i y_i & x_i^3 & x_i^2 y_i & x_i y_i^2 \\ y_i & x_i y_i & y_i^2 & x_i^2 y_i & x_i y_i^2 & y_i^3 \\ x_i^2 & x_i^3 & x_i^2 y_i & x_i^4 & x_i^3 y_i & x_i^2 y_i^2 \\ x_i y_i & x_i^2 y_i & x_i y_i^2 & x_i^3 y_i & x_i^2 y_i^2 & x_i y_i^3 \\ y_i^2 & x_i y_i^2 & y_i^3 & x_i^2 y_i^2 & x_i y_i^3 & y_i^4 \end{pmatrix} \begin{pmatrix} s \\ r_x \\ r_y \\ q_{xx} \\ q_{xy} \\ q_{yy} \end{pmatrix} = \sum_{i=0}^n \theta(d_i) \begin{pmatrix} 1 \\ x_i \\ y_i \\ x_i^2 \\ x_i y_i \\ y_i^2 \end{pmatrix} z_i$$

$\theta : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ (non-linear) weighting function

$d_i = \left\| \begin{pmatrix} x_i \\ y_i \end{pmatrix} - \begin{pmatrix} x_j \\ y_j \end{pmatrix} \right\|$ distance in parameter space



Exercise 2 Practice Part

x 1 - Weighted Least Squares II

What do we get after solving the weighted least squares system?

The parameter vector $(s, r_x, r_y, q_{xx}, q_{xy}, q_{yy})$ that belongs to a point in space (x_j, y_j)

The parameters represent the coefficients of the **local** bivariate quadratic polynomial at (x_j, y_j)



Exercise 2 Practice Part

x 1 - Moving Least Squares

Moving Least Squares:

Apply Weighted Least Squares for every point $\mathbf{x}_j$ in parameter domain

This yields a "local" function f_j at that point

The function value $z_j = f_j(\mathbf{x}_j)$ is the remaining coordinate for that point

Doing this for all points in space leads to a (smooth) surface

Of course, we can only evaluate the surface at finitely many (discrete) parameter values



Exercise 2 Practice Part

Task 1 - Wendland's Weighting Function

Wendland's weighting function (C^2 -continuous):

$$\theta(d) = \begin{cases} (1 - \frac{d}{h})^4 (\frac{4d}{h} + 1) & d \leq h \\ 0 & d > h \end{cases} \quad \leftarrow \text{local support}$$

Local support: From a specific distance on, all weights become zero

Don't need to consider too distant points for the computation

The weighting and "cut-off" are controlled with the hyperparameter h

Should be adjustable via the GUI

By choosing a non-linear weighting function, we can also choose simpler basis functions (next exercise)



Exercise 2 Practice Part x 2

Use the $m \times n$ grid points as control points for a Bézier tensor product surface with $km \times kn$ points

You should be able to adjust k in the GUI

Use de Casteljau's algorithm to create the points on the surface

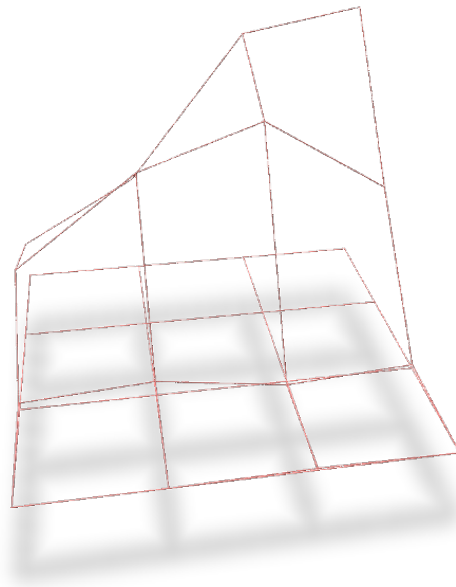
The control mesh is a regular grid in the x - y -plane

$\Rightarrow$ think of an optimization

Visualize the surface as a surface mesh

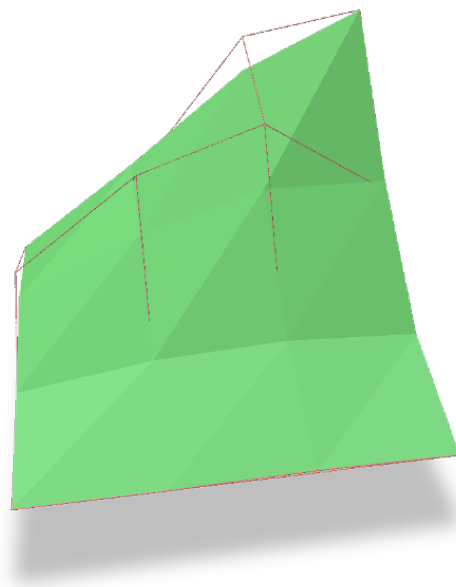


Exercise 2 Practice Part x 2 visualized

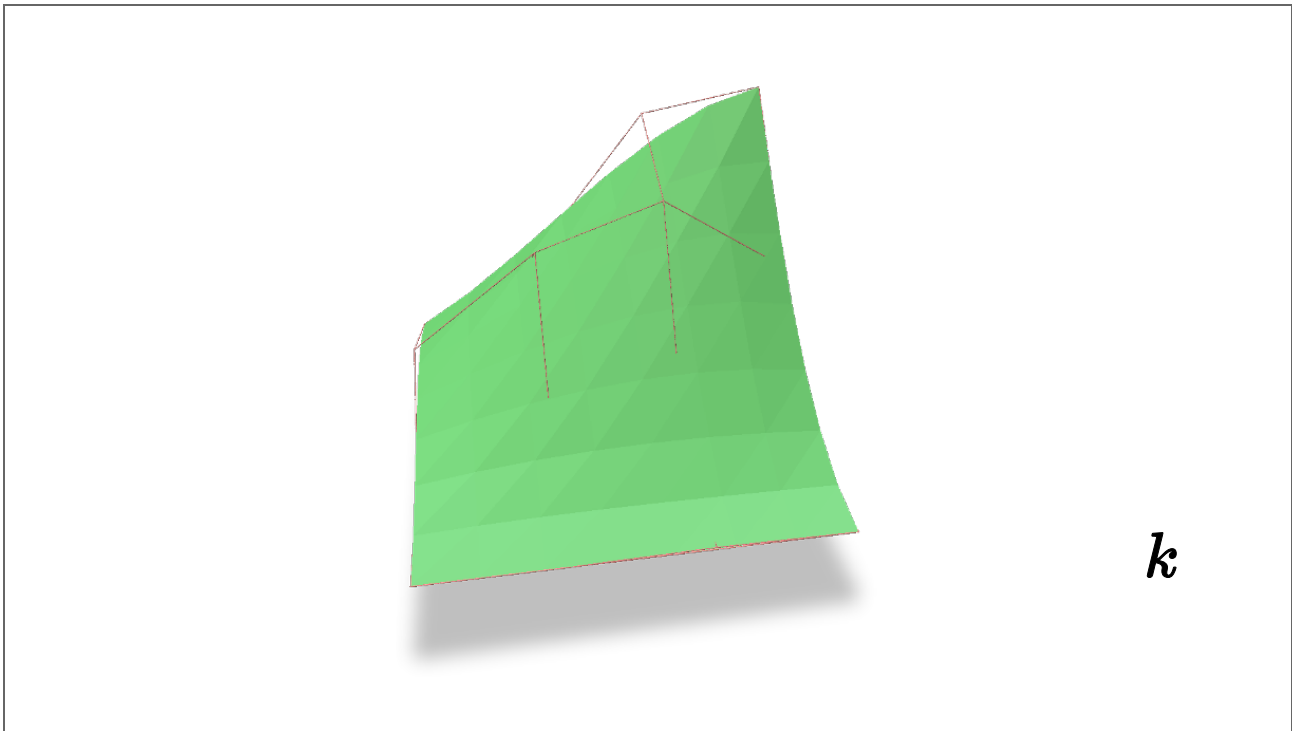


Resume presentation

Bézier control grid



Bézier surface



Bézier surface with higher resolution (larger k)



Exercise 2

x 3

Compute the normals of the Bézier surface

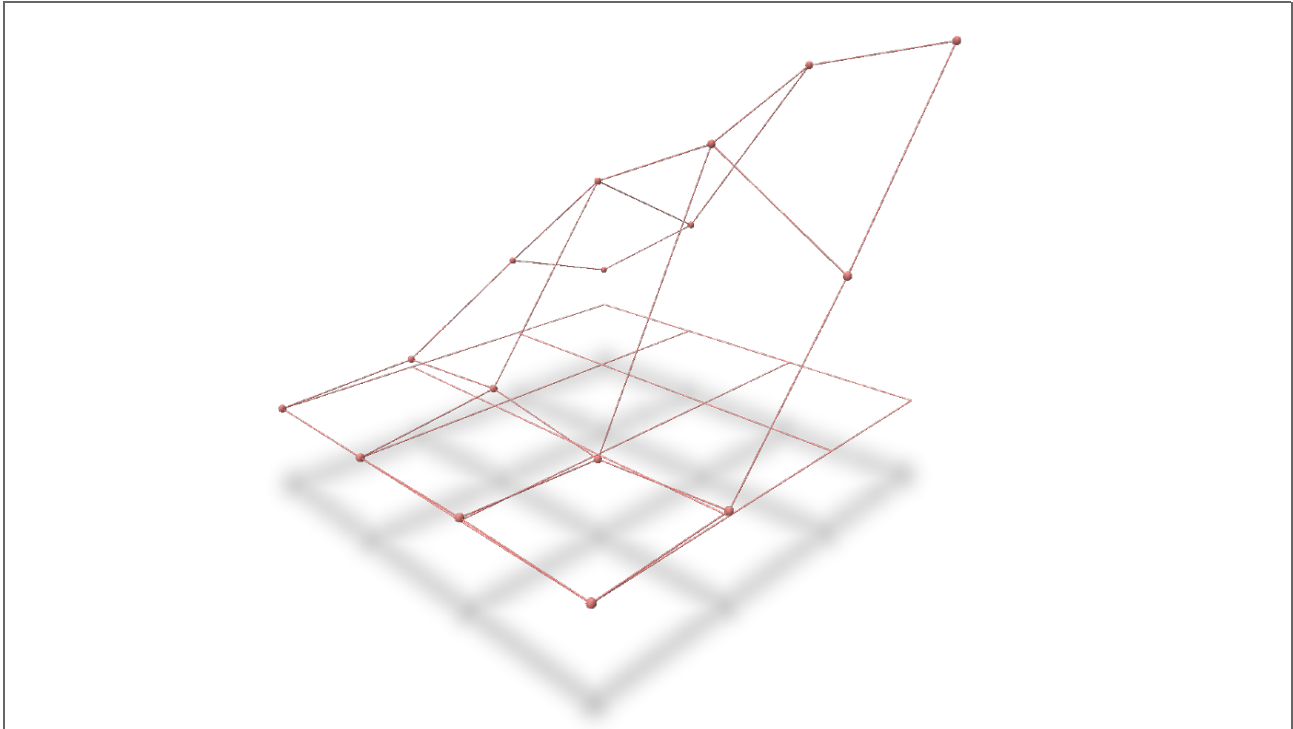
Compute the tangents with de Casteljau's algorithm

How to get the normal from the tangents?

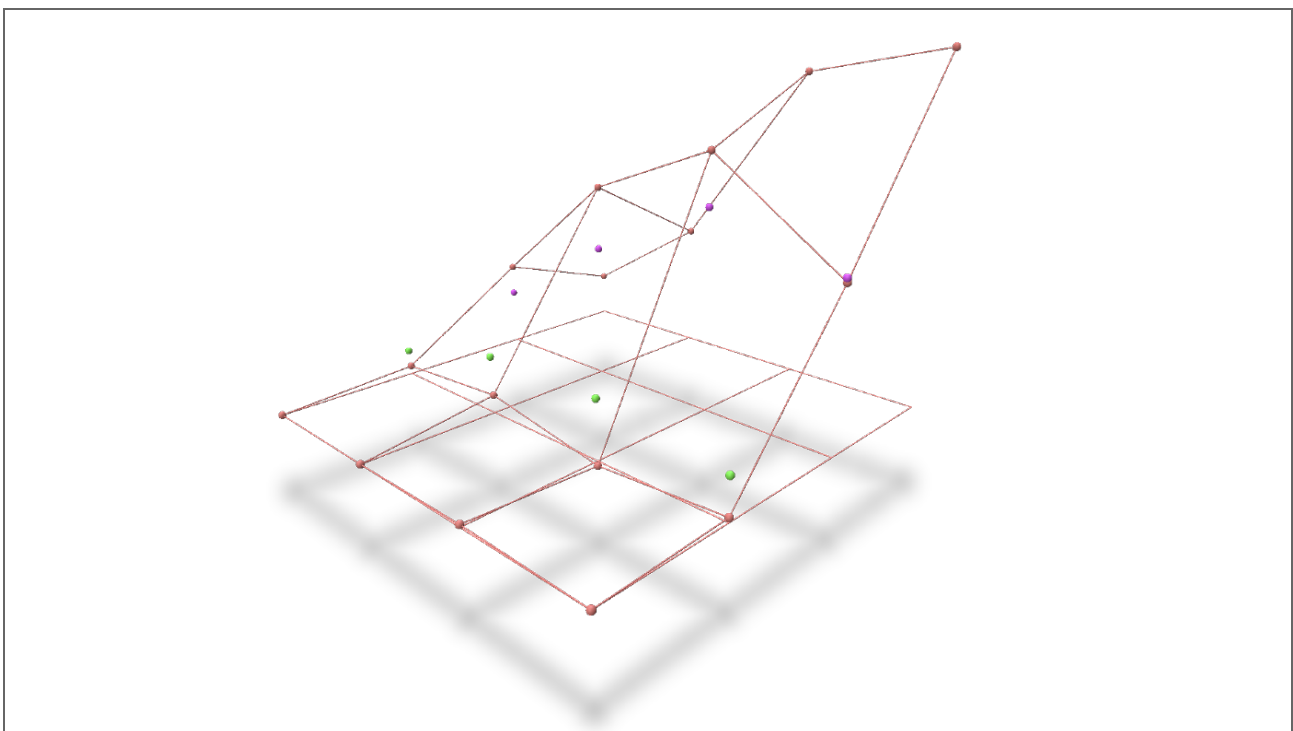
Visualize the normals

Exercise 2

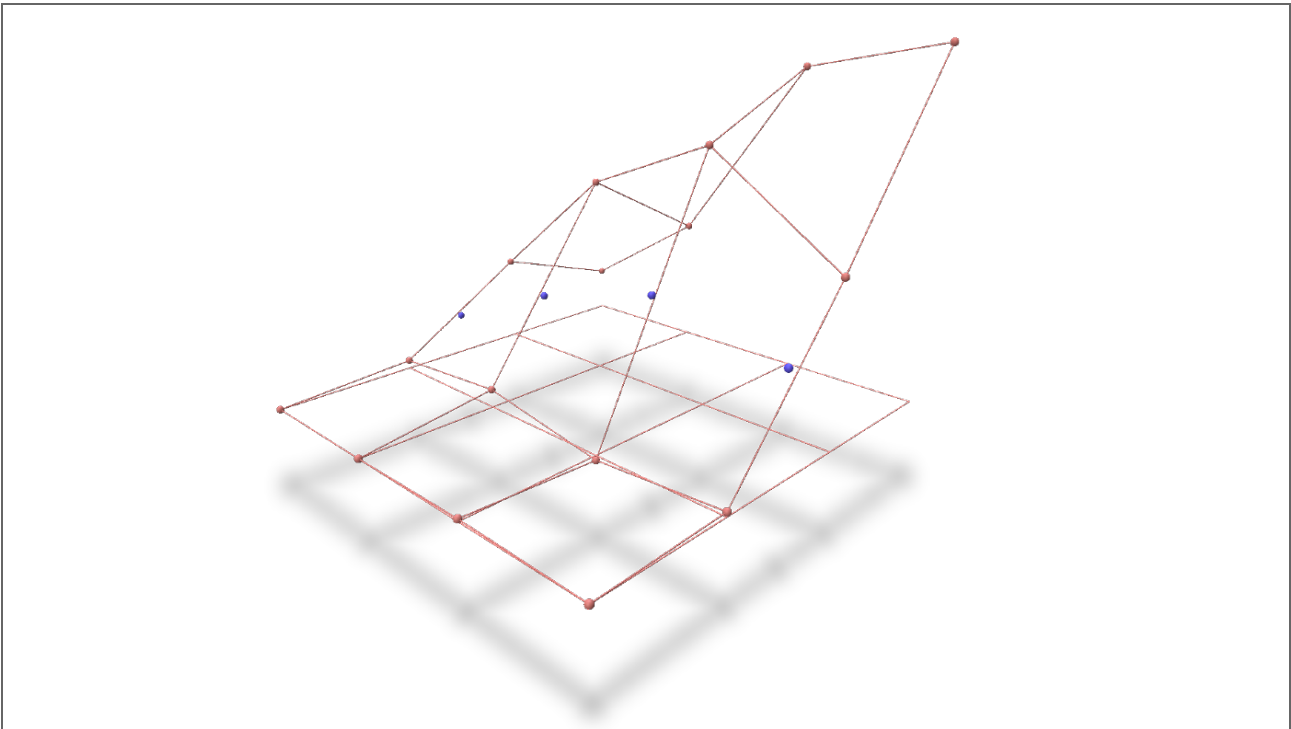
x 3 visualized



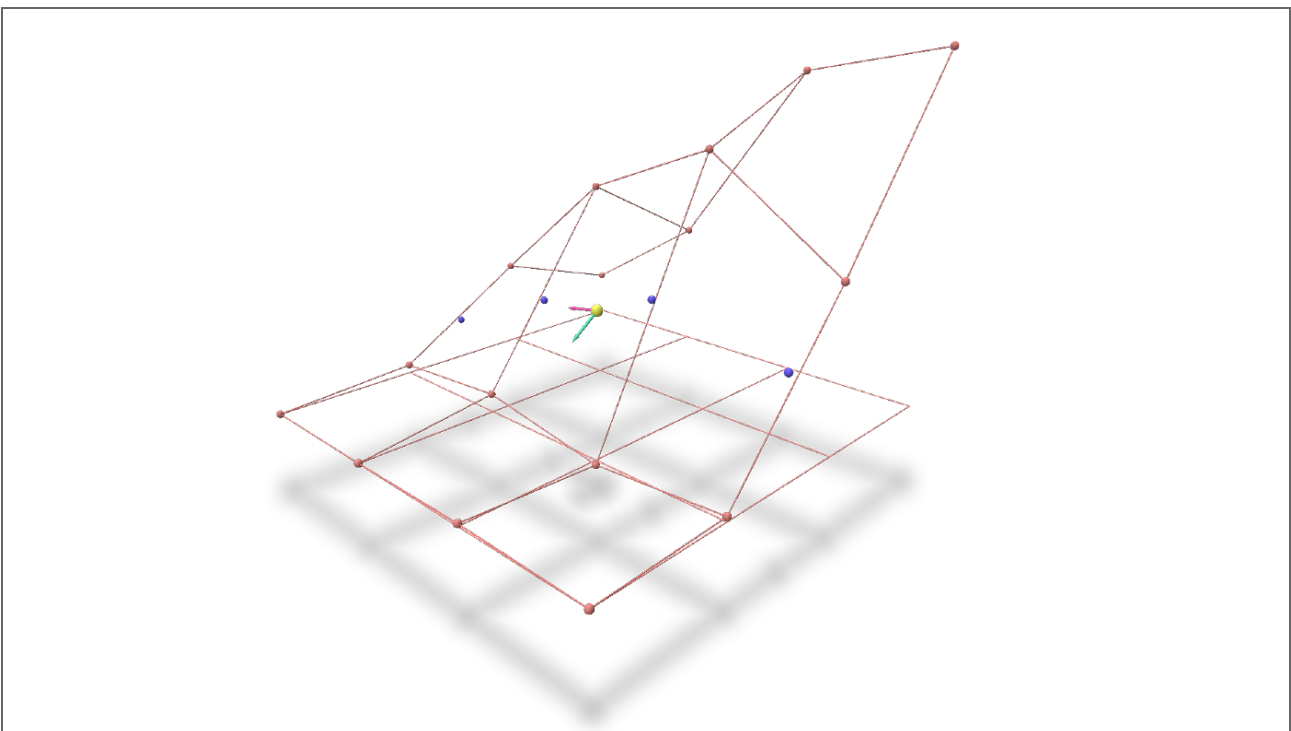
Bézier control grid



De Casteljau (second last step)



De Casteljau (last step)



De Casteljau (result with tangents)



Exercise 2

x 4

Compute an MLS surface with resolution of $km \times kn$
 k adjustable as in task 2

Visualize the MLS surface as a surface mesh

Should be easy if task 1 is already done



Exercise 2

x 5

Compute partial derivatives (tangents) of the local WLS polynomial

From them, compute normals

Visualize the normals

Should be solved along theory task 5



Exercise 2

Speaker notes